

奠基全球的科技巨擘



1988年

廣達電腦成立，開啟科技新局



Fortune 500

榮登全球五百大企業之列

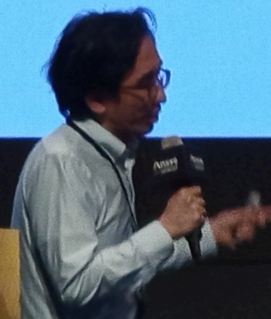
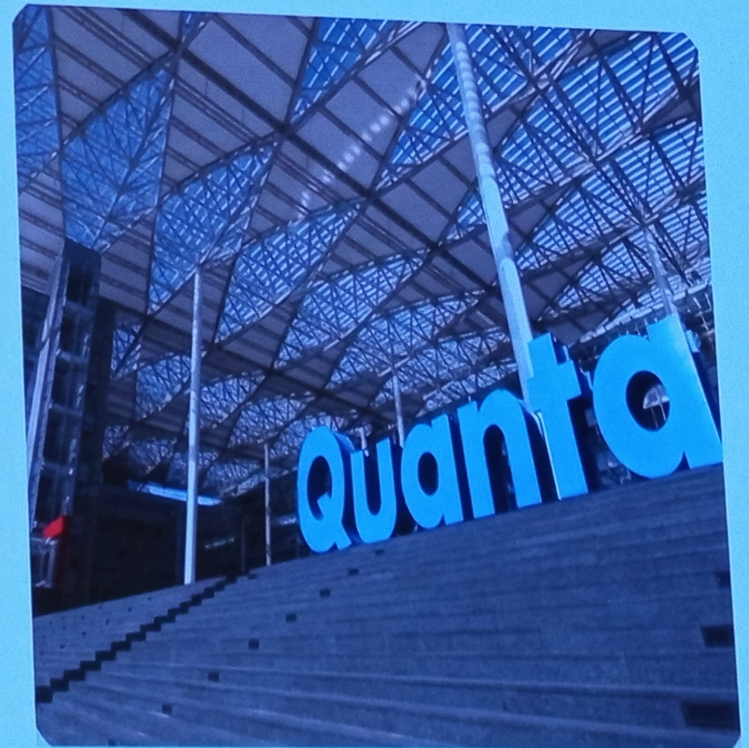
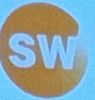


No.1

全球筆記型電腦專業研發
設計製造的領導廠商

大綱

- 廣達電腦簡介
- Sherlock 模擬理論與失效機制
- MCU元件可靠度評估
- 變壓器元件可靠度評估



驅動世界的創新引擎

2019-2022 連續四年獲得

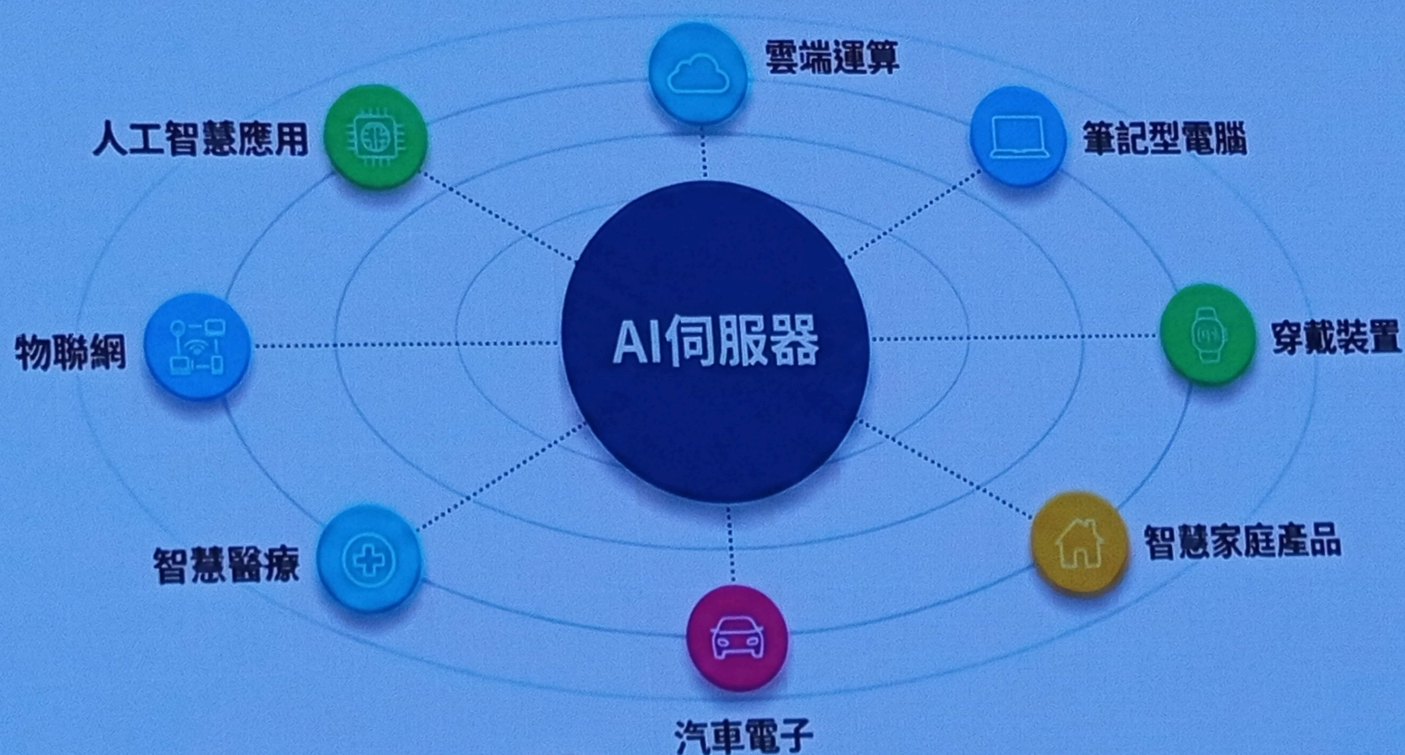
全球百大創新機構獎

科睿唯安 Clarivate



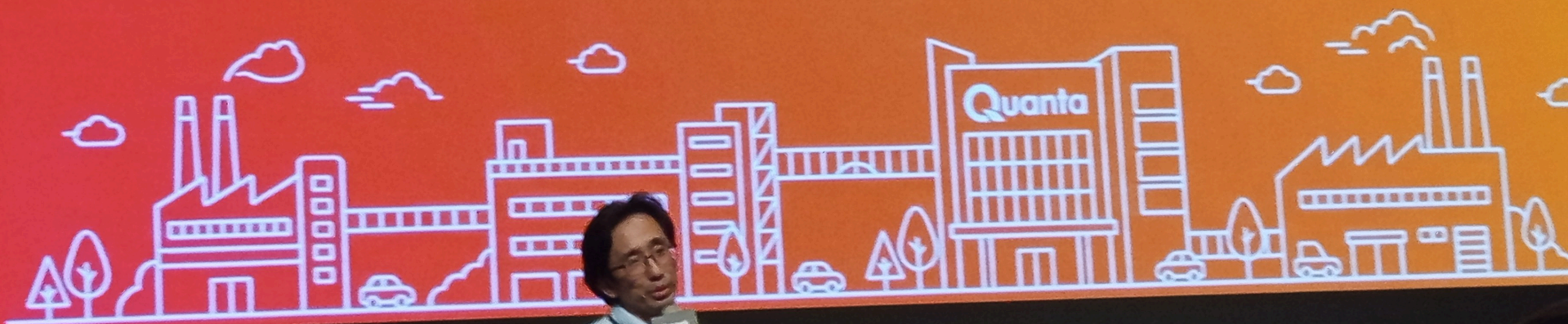
超越筆電的無限未來

積極拓展產業領域，開創商機並整合相關資源進行佈局

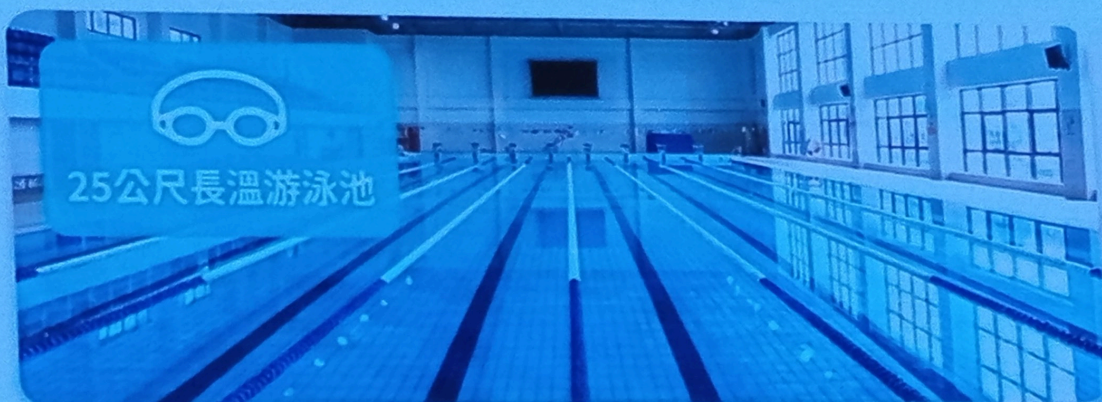


廣達好生活：工作與生活的完美共鳴

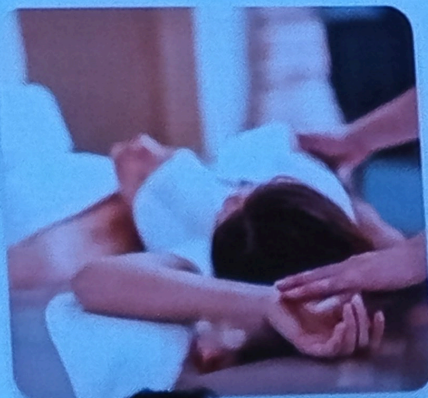
走進廣達，體驗全方位的幸福職場



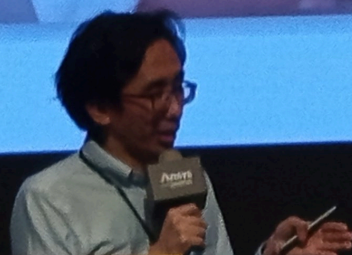
員工休閒中心：釋放壓力的頂級聚落



球類天地: 籃球、羽毛球、桌球、撞球

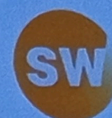


舒壓專區: 盲人按摩理療、KTV、閱覽室



Sherlock 模擬理論與失效機制

前言：壽命分析的重要性

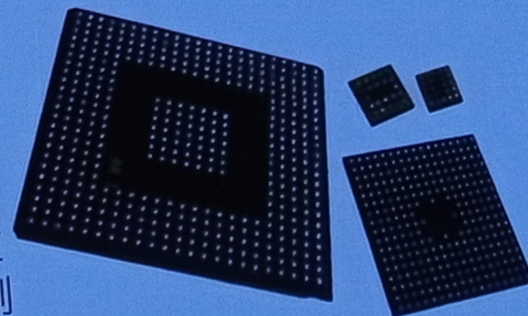


錫球應用的重要性

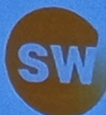
- 1. 電氣連接：**錫球提供電子元件與基板之間的電氣連接，確保訊號傳輸的穩定性和可靠性。
- 2. 機械支撐：**錫球同時負責支撐元件，保持其在基板上的物理位置，並吸收機械應力。

壽命評估的重要性

- 1. 可靠性評估：**由於錫球在承受熱衝擊、機械應力和電氣應力時容易發生疲勞和失效，因此對錫球壽命進行分析至關重要。壽命分析可以幫助預測元件的使用壽命，避免過早失效。
- 2. 失效預測與改善設計：**通過壽命分析，工程師可以預測錫球在不同應用條件下的失效模式，從而針對性地優化封裝設計和材料選擇，提升產品的耐用性和可靠性。
- 3. 降低維護成本與風險：**如果錫球的壽命較短，可能導致整個電子系統的故障，造成維護和更換成本的增加。通過提前進行壽命分析，可以降低這些風險，延長產品壽命。



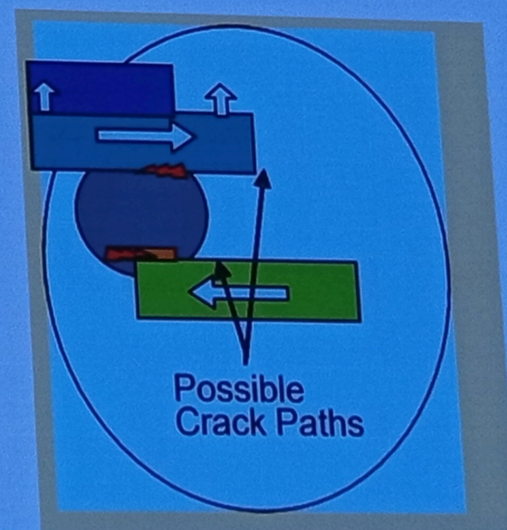
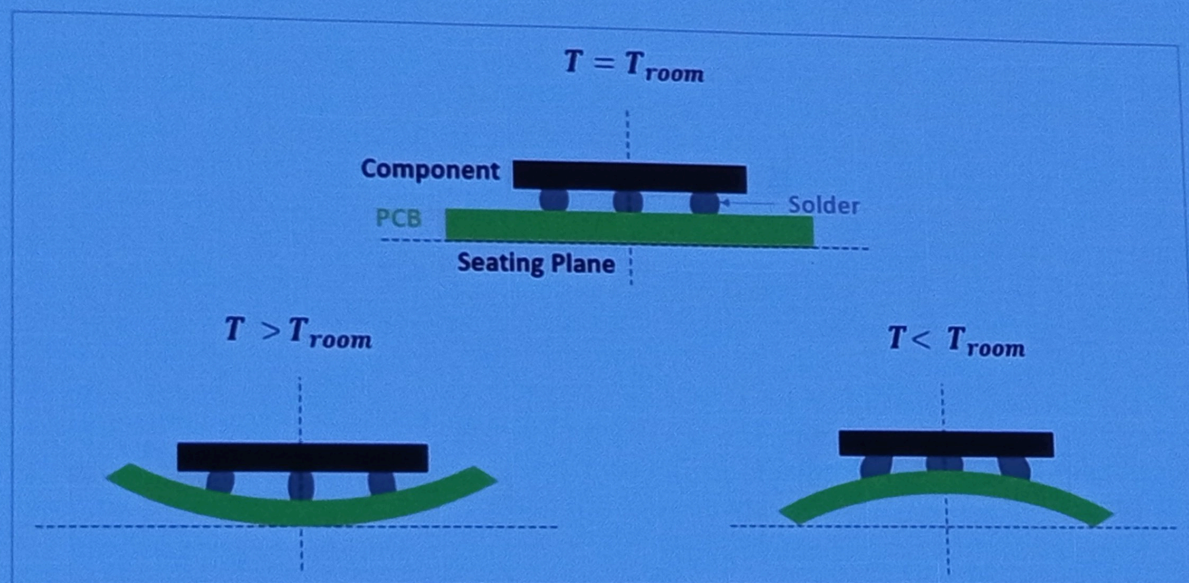
熱衝擊(1)



What is Thermo-Mechanical Solder Joint Fatigue?

因膨脹或收縮的差異會使焊點承受反覆的剪切載荷。通常，這種剪切載荷或應力足以引起焊點的非彈性應變。重複的溫度循環（例如日夜溫差變化和電源開關操作）會導致應變累積，從而損壞焊點。隨著每次額外的溫度循環，這種損壞會增加，最終導致裂紋擴展，並引發焊點的失效

Figure 2.1: Effect of Thermal Cycling

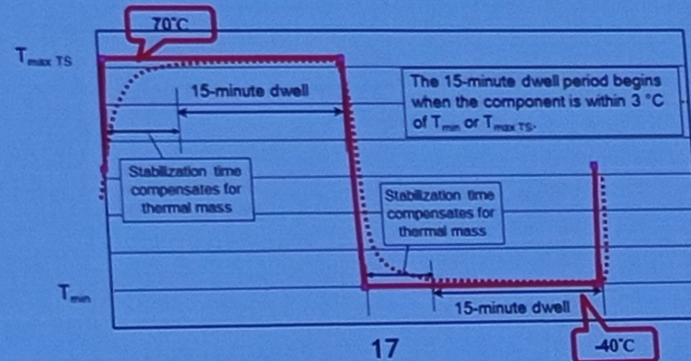


熱衝擊(2)

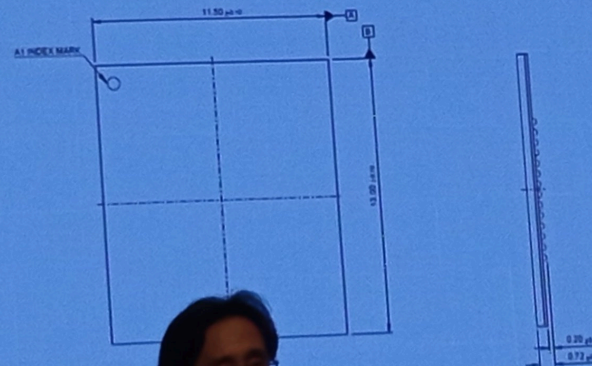


What are the drivers for Thermo-Mechanical Solder Joint Fatigue?

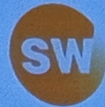
1. Temperature Profile: Maximum and minimum temperatures, dwell times at maximum and minimum temperature



2. Component Design: Such as size, and other factors



Research Methodology :Application of Sherlock



Define Material Parameters

From Library: In addition to allowing users to define packaging parameters based on datasheets, Sherlock also has a built-in library over one million .

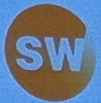
The screenshot displays the Sherlock software interface. At the top, there are search filters: 'Use AVL: YES', 'Use Wizard: YES', 'Manufacturer: MARVELL', and a list of criteria including MACRONIX, MAG LAYERS, MAGNETICS INC, MAIDA DEVELOPMENT COMPANY, MARKI MICROWAVE, MARVELL, MATSUSHITA ELECTRONIC, and MAXIM. A 'Customize' button is visible. Below the filters, a table titled 'Search Results (18)' lists various components. The table has columns for Manufacturer, Part Number, Part Type, and Part Description. The results include components from MARVELL, such as integrated gigabit ethernet transceivers, digital interfaces, and ethernet transceivers.

Manufacturer	Part Number	Part Type	Part Description
MARVELL	88E1111-B2-ND2C000	IC	Integrated 10/100/1000 Ultra Gigabit Ethernet Transceiver, Commercial Grade, 96-Pin QFN...
MARVELL	88E1112-C2-NNC1C000	IC	IC, DIGITAL, INTERFACE, INTEGRATED 10/100/1000 GIGABIT ETHERNET TRANSCIVER, 2.5V/...
MARVELL	88E116RA0-NNC1C000	IC	Gigabit Ethernet Transceiver
MARVELL	88E1510PXX-NNB21000	IC	IC, DIGITAL, INTERFACE, 88E1510P, ETHERNET TRANSCIVER, 1.8V/3.3V, QFN48, SMD
MARVELL	88E1512-A0-NNP21000	IC	IC ETH TRANC PHY SGMII QFN 56
MARVELL	88E1512PXX-NNP21000	IC	IC, ENET XCVR, 10/100/1000 Mbps, QFN-56
MARVELL	88E1514-A0-NNP2C000	IC	IC, 10/100/1000 MBPS ENERGY EFFICIENT ETHERNET TRCVR
MARVELL	88E1514-XX-NNP2C000	IC	IC, 10/100/1000 MBPS ENERGY EFFICIENT ETHERNET TRCVR
MARVELL	88E1548PA0-BAM21000	IC	IC, DIGITAL, QUAD PORT, 10/100/1000Mbps/s ENERGY EFFICIENT ETHERNET TRANSCIVE...
MARVELL	88E3018-A2-NNC11000	IC	IC, 10/100 FAST ETHERNET TXXR, 2.5V MII/RGMII INTERFACE, ITR, ROHS, HT-QFN64
MARVELL	88EA1512-B2-NNP2A000	IC	IC 10/100/1000 ETH Transceiver 56QFN
MARVELL	88EA1512B2-NNP2A000	IC	1000Mbps Ethernet Transceiver
MARVELL	88Q2112-A2-NYD2A000	IC	Ethernet ICs Fast Ethernet PHY
MARVELL	88Q2112-XX-NYD2A000-P...	IC	
MARVELL	88Q2221-B0-NYA2A0G2	IC	
MARVELL	88Q6113-B2-BYZZA000	IC	
MARVELL	88X2222PA0-BKP2C000	IC	IC DUAL MULTI-SPEED ETHERNET TRCVR W/EDC TECH
MARVELL	340-88E1512A0NNP2100	IC	Ethernet ICs Single-port EEE GE PHY with SGMII in 56-pin QFN package Industrial Temp...

The screenshot displays the material parameters section of the Sherlock software. On the left, there is a vertical list of categories: ID, Package, Location, Ball, Pad, Solder, Electrical, Thermal, Die, Flag, Test, Semiconductor Wearout, and User. The 'Package' category is selected. The main area shows various parameters for a package, including Package Name, Package Type, Package Mount, Package Units, Package Length, Package Width, Overmold Thickness, Laminate Thickness, Model Part, Corner Shape, Corner Radius, Corner Face, Substrate Material, and Overmold Material. Each parameter has a dropdown menu or a text input field, and a 'User' button is visible next to several parameters.

Parameter	Value	Source
Package Name	BGA356-MARVELL-88Q5072-0_55	0105_MB.csv
Package Type	BGA	0105_MB.csv
Package Mount	SMT	0105_MB.csv
Package Units	MM	0105_MB.csv
Package Length	17	User
Package Width	17	0105_MB.csv
Overmold Thickness	0.7	0105_MB.csv
Laminate Thickness	0.36	0105_MB.csv
Model Part	ENABLED	0105_MB.csv
Corner Shape	SQUARE	0105_MB.csv
Corner Radius	0	0105_MB.csv
Corner Face	TOP_BOTTOM	0105_MB.csv
Substrate Material	LAMINATE-BGA	User
Overmold Material	OVERMOLD-BGA	User

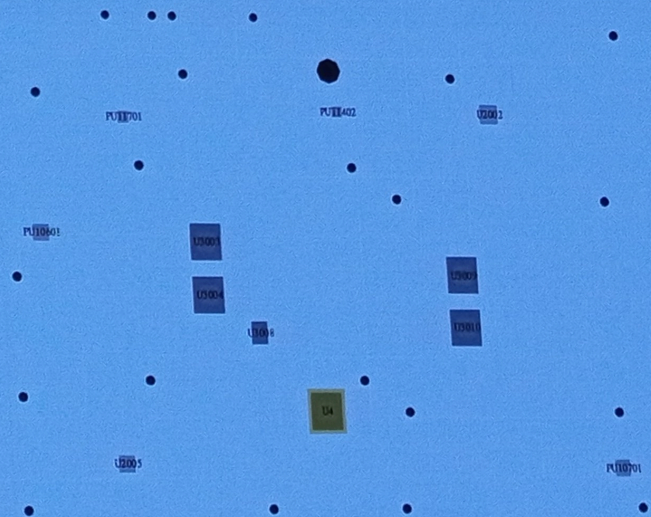
Research Methodology :Application of Sherlock



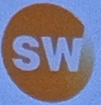
Sherlock supports importing ECAD files such as ODB++

Parts Listing (22)

Ref Des	Part Number	Part Type	Package	Location
U3008	BGA24-MXC-MX2SL51245GXD-0_5S	IC	SMT BGA24-MXC-MX2SL51245GXD-0_5S	BOTTOM
U7	BGA66-RENEAS-RAA271005-0_5S	IC	SMT BGA66-RENEAS-RAA271005-0_5S	TOP
U4	BGA153-SANDISK-SDIN7DP2-BG-0_5S	IC	SMT BGA153-SANDISK-SDIN7DP2-BG-0_5S	BOTTOM
U3003	BGA153-WESTERN-SDINBDA6-128G-0_5S	IC	SMT BGA153-WESTERN-SDINBDA6-128G-0_5S	BOTTOM
U3004	BGA153-WESTERN-SDINBDA6-128G-0_5S	IC	SMT BGA153-WESTERN-SDINBDA6-128G-0_5S	BOTTOM
U3009	BGA153-WESTERN-SDINBDA6-128G-0_5S	IC	SMT BGA153-WESTERN-SDINBDA6-128G-0_5S	BOTTOM
U3010	BGA153-WESTERN-SDINBDA6-128G-0_5S	IC	SMT BGA153-WESTERN-SDINBDA6-128G-0_5S	BOTTOM
U3	BGA200-MICRON-MTS3D512M32D2DS-0_5S	IC	SMT BGA200-MICRON-MTS3D512M32D2DS-0_5S	TOP
U2003	BGA288-MARVELL-88Q5152	IC	SMT BGA288-MARVELL-88Q5152	TOP
U2004	BGA356-MARVELL-88Q5072-0_5S	IC	SMT BGA356-MARVELL-88Q5072-0_5S	TOP
U1	BGA780-RENEAS-R8A779F5	IC	SMT BGA780-RENEAS-R8A779F5	TOP
U3001	BGA1573-QUALCOMM-QAM8650P	IC	SMT BGA1573-QUALCOMM-QAM8650P	TOP
U3002	BGA1573-QUALCOMM-QAM8650P	IC	SMT BGA1573-QUALCOMM-QAM8650P	TOP
PU11402	QFN20-4X4-S-21P-MAX17633	IC	SMT QFN-20 (MO-220WGGD-1)	BOTTOM
PU11701	QFN22-4_25X4_25-S-20P	IC	SMT QFN-22 (MO-248UHC)	BOTTOM
U6	QFN24-4X4-S-25P-SP49V60	IC	SMT QFN-24 (MO-220VGGD-9)	TOP
PU10601	QFN40-6X6-S-41P-LM5143	IC	SMT QFN-40 (MO-220VJJD-6)	BOTTOM
PU10701	QFN40-6X6-S-41P-LM5143	IC	SMT QFN-40 (MO-220VJJD-6)	BOTTOM



Research Methodology :Application of Sherlock



Definition of PCB Stackup

Stackup Properties

The following board properties are based on the currently defined board outline and the individual layer properties shown below:

Board Dimension: 242.98 x 190.07 mm [9.5661 x 7.4831 in]
Board Thickness: 1.586 mm [62.4 mil]
Density: 3.1166 g/cc
Conductor Layers: 10

Sherlock can calculate the equivalent E value and CTE

CTExy: 15.627 ppm/C
CTEz: 35.943 ppm/C
Exy: 39,151 MPa
Ez: 11,519 MPa

Board Weight: 223.4 grams
Total Part Weight: 38.79 grams
Mount Point Weight: 0 grams
Fixture Weight: 0 grams

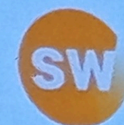
Stackup Layers

Double-click any row to edit the properties for that layer or select one or more rows and press the **Edit** button above to edit properties for the selected layers. Use the **Generate Stackup** button to replace all layers using a given PCB thickness and default layer properties.

Layer	Type	Material	Construction	Thickness	Density	CTExy	CTEz	Exy	Ez
1	SIGNAL	COPPER (60.9%) / RM_L01		1.8 mil	6.1630	15.410	15.410	79,836	79,836
2	Laminate	EM-370Z / Resin Content (67.0%)	1080	3.03 mil	1.7653	16.447	44.217	19,580	8,621
3	SIGNAL	COPPER (93.3%) / RM_L03		1.8 mil	8.4310	17.225	17.225	107,317	107,317
4	Laminate	EM-370Z / Resin Content (69.0%)	1080	3.13 mil	1.7494	17.193	44.980	18,740	8,476
5	SIGNAL	COPPER (67.1%) / RM_L05		1.2 mil	6.5970	15.758	15.758	85,095	85,095
6	Laminate	EM-370Z / Resin Content (56.0%)	2116x2	10 mil	1.8524	13.270	40.020	24,203	9,521
7	SIGNAL	COPPER (94.6%) / RM_L07		1.2 mil	8.5220	17.298	17.298	108,420	108,420
8	Laminate	EM-370Z / Resin Content (69.0%)	1080	2.87 mil	1.7494	17.193	44.980	18,740	8,476
9	SIGNAL	COPPER (83.3%) / RM_L09		1.2 mil	7.7310	16.665	16.665	98,835	98,835
10	Laminate	EM-370Z / Resin Content (56.0%)	2116x2	10 mil	1.8524	13.270	40.020	24,203	9,521
11	SIGNAL	COPPER (83.3%) / RM_L11		1.2 mil	7.7310	16.665	16.665	98,835	98,835
12	Laminate	EM-370Z / Resin Content (69.0%)	1080	2.87 mil	1.7494	17.193	44.980	18,740	8,476
13	SIGNAL	COPPER (94.6%) / RM_L13		1.2 mil	8.5220	17.298	17.298	108,420	108,420
14	Laminate	EM-370Z / Resin Content (56.0%)	2116x2	10 mil	1.8524	13.270	40.020	24,203	9,521
15	SIGNAL	COPPER (67.1%) / RM_L15		1.2 mil	6.5970	15.758	15.758	85,095	85,095
16	Laminate	EM-370Z / Resin Content (69.0%)	1080	3.13 mil	1.7494	17.193	44.980	18,740	8,476
17	SIGNAL	COPPER (93.3%) / RM_L17		1.8 mil	8.4310	17.225	17.225	107,317	107,317
18	Laminate	EM-370Z / Resin Content (67.0%)		3.03 mil	1.7653	16.447	44.217	19,580	8,621
19	SIGNAL	COPPER (60.9%) / RM_L19		1.8 mil	6.1630	15.410	15.410	79,836	79,836

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試條件



Vibration

c) M₁ 类、N₁ 类车辆,带悬置的电机、集成式驱动电机系统,随机振动条件如图4和表8所示。

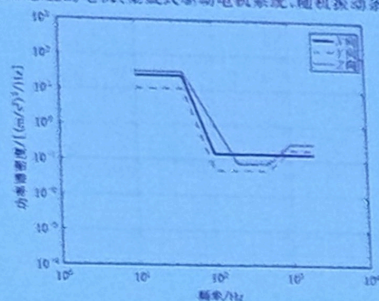


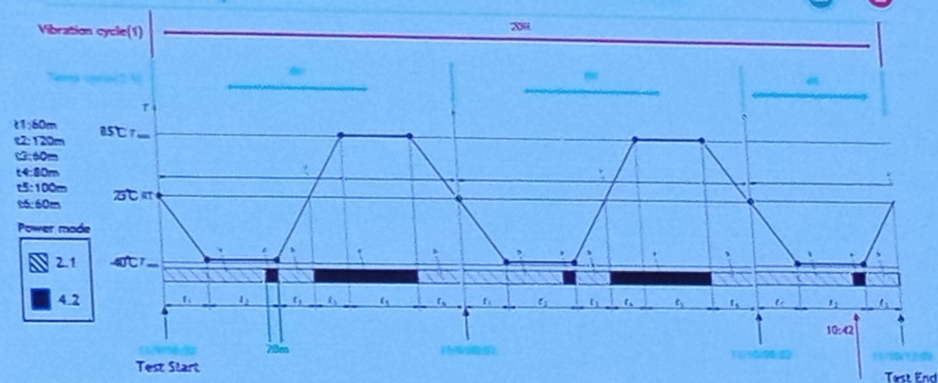
图4 M₁ 类、N₁ 类车辆,带悬置的电机、集成式驱动电机系统,随机振动条件

表8 M₁ 类、N₁ 类车辆,带悬置的电机、集成式驱动电机系统,随机振动条件

频率/Hz	PSD / [(m/s²)²/Hz]		
	X 方向	Y 方向	Z 方向
10	25	11	33
40	25	11	33
100	0.15	0.05	—
200	—	—	0.05
500	—	0.05	0.05
1 000	—	0.2	0.3
2 000	0.15	0.2	0.3
RMS/(m/s²)	35.4	26.4	43.3
试验时长	20 h	20 h	20 h

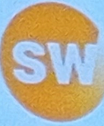
Thermal

Test Fail Sequence



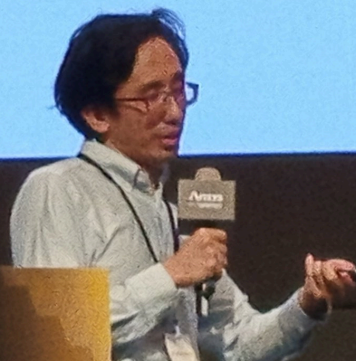
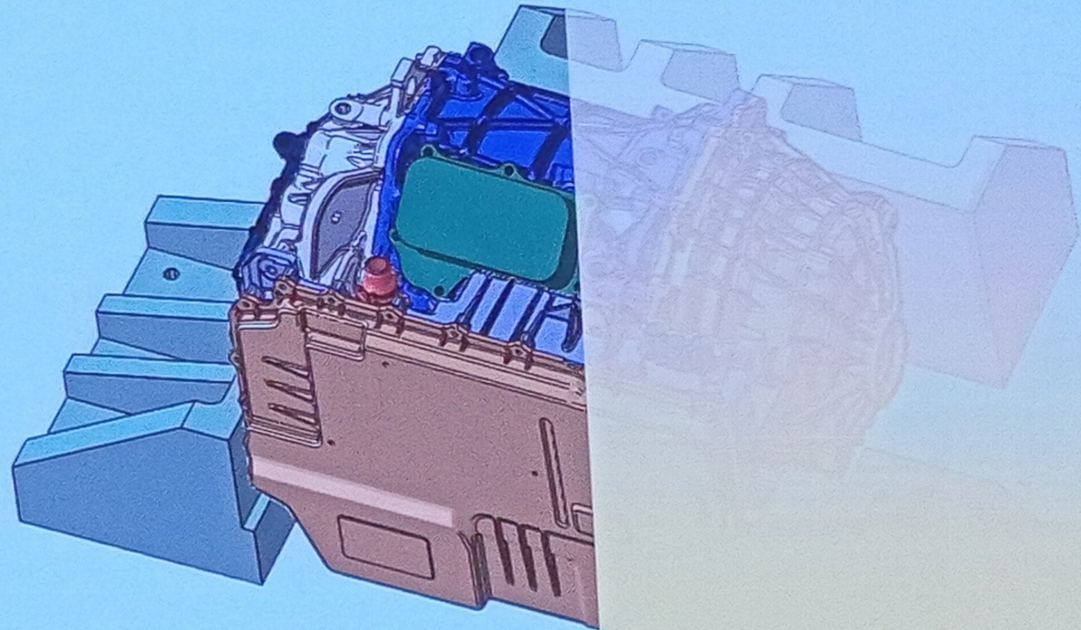
Test duration : 20hrs per axis

完整部件(包含馬達)

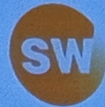


Weight of motor:84.7kg
Weight of Fixture:40.9kg
...

Fixture



Research Methodology :Application of Sherlock



Define Material Parameters

From Library: In addition to allowing users to define packaging parameters based on datasheets, Sherlock also has a built-in library over one million .

The screenshot displays the Sherlock software interface. At the top, there are search filters: 'Use AVL' (YES), 'Use Wizard' (YES), 'Manufacturer' (MARVELL), and 'Part Number' (MACRONIX). Below these, a list of manufacturers is shown, including MAG LAYERS, MAGNETICS INC, MAIDA DEVELOPMENT COMPANY, MARKI MICROWAVE, MARVELL, MATSUSHITA ELECTRONIC, and MAXIM. A 'Customize' button is also visible.

Below the filters, a table titled 'Search Results (18)' lists various components. The table has four columns: Manufacturer, Part Number, Part Type, and Part Description.

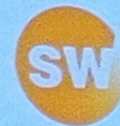
Manufacturer	Part Number	Part Type	Part Description
MARVELL	88E1111-B2-ND2C000	IC	Integrated 10/100/1000 Ultra Gigabit Ethernet Transceiver, Commercial Grade, 96-Pin QFN...
MARVELL	88E1112-C2-NNC1C000	IC	IC, DIGITAL, INTERFACE, INTEGRATED 10/100/1000 GIGABIT ETHERNET TRANSCIVER, 2.5V/...
MARVELL	88E116RA0-NNC1C000	IC	Gigabit Ethernet Transceiver
MARVELL	88E1510PXX-NNB21000	IC	IC, DIGITAL, INTERFACE, 88E1510P, ETHERNET TRANSCIVER, 1.8V/3.3V, QFN48, SMD
MARVELL	88E1512-A0-NNP21000	IC	IC ETH TRANC PHY SGMII QFN 56
MARVELL	88E1512PXX-NNP21000	IC	IC, ENET XCVR, 10/100/1000 Mbps, QFN-56
MARVELL	88E1514-A0-NNP2C000	IC	IC, 10/100/1000 MBPS ENERGY EFFICIENT ETHERNET TRCVR
MARVELL	88E1514-XX-NNP2C000	IC	IC, 10/100/1000 MBPS ENERGY EFFICIENT ETHERNET TRCVR
MARVELL	88E1548PA0-BAM21000	IC	IC, DIGITAL, QUAD PORT, 10/100/1000Mbps/s ENERGY EFFICIENT ETHERNET TRANSCIVE...
MARVELL	88E3018-A2-NNC11000	IC	IC, 10/100 FAST ETHERNET TXX, 2.5V MII/RGMII INTERFACE, ITR, ROHS, HT-QFN64
MARVELL	88EA1512-B2-NNP2A000	IC	IC 10/100/1000 ETH Transceiver 56QFN
MARVELL	88EA1512B2-NNP2A000	IC	1000Mbps Ethernet Transceiver
MARVELL	88Q2112-A2-NYD2A000	IC	Ethernet ICs Fast Ethernet PHY
MARVELL	88Q2112-XX-NYD2A000-P...	IC	
MARVELL	88Q2221-B0-NYA2A0G2	IC	
MARVELL	88Q6113-B2-BYZZA000	IC	
MARVELL	88X2222PA0-BKP2C000	IC	IC DUAL MULTI-SPEED ETHERNET TRCVR W/EDC TECH
MARVELL	340-88E1512A0NNP2100	IC	Ethernet ICs Single-port EEE GE PHY with SGMII in 56-pin QFN package Industrial Temp...

The screenshot displays the Sherlock software interface for defining material parameters. On the left, there is a vertical menu with options: ID, Package, Location, Ball, Pad, Solder, Electrical, Thermal, Die, Flag, Test, Semiconductor Wearout, and User. The 'Package' option is selected.

On the right, there are various input fields for defining the package parameters:

- Package Name: BGA356-MARVELL-88Q5072-0_55
- Package Type: BGA
- Package Mount: SMT
- Package Units: MM
- Package Length: 17
- Package Width: 17
- Overmold Thickness: 0.7
- Laminate Thickness: 0.36
- Model Part: ENABLED
- Corner Shape: SQUARE
- Corner Radius: 0
- Corner Face: TOP_BOTTOM
- Substrate Material: LAMINATE-BGA
- Overmold Material: OVERMOLD-BGA

完整部件(包含馬達)



Weight of motor:84.7kg
Weight of Fixture:40.9kg

Fixture

